

MATHEMATICS**Maximum Marks: 80****Time Allowed: Three Hours***Reading Time: Additional Fifteen Minutes***Instructions to Candidates**

1. You are allowed an additional **fifteen minutes** for **only** reading the paper.
2. You must **NOT** start writing during reading time.
3. The question paper has **10** printed pages.
4. It consists of **twenty** questions and **four** sections: Sections A, B, C and D.
5. All questions are compulsory.
6. Section A consists of very short answer questions of 1 mark each.
7. Section B consists of short answer questions of 2 marks each.
8. Section C consists of moderately long answer questions of 3 marks each.
9. Section D consists of long answer questions of 5 marks each.
10. Internal choices have been provided in **three** questions, each in Sections B, C and D.
11. While attempting Multiple Choice Questions in Section A, you are required to write only **ONE** option as the answer.
12. The intended marks for questions or parts of questions are given in the brackets [].
13. All workings, including rough work, should be done on the same page as, and adjacent to, the rest of the answer.
14. Mathematical tables and graph papers are provided.

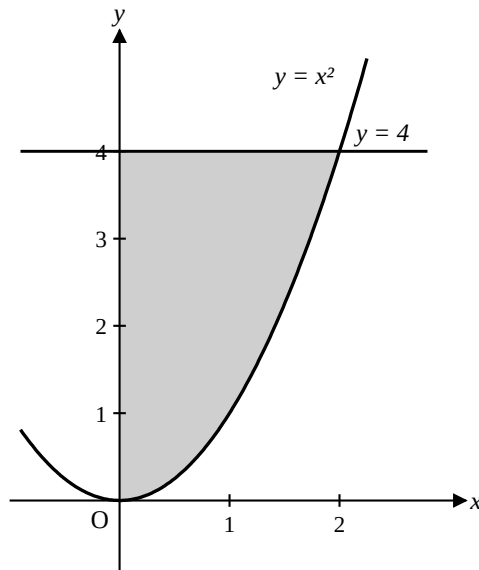
Instruction to Supervising Examiner*Kindly read aloud the instructions given above to all the candidates present in the examination hall.***SECTION A – 20 MARKS****Question 1**

In subparts (i) to (xvii) choose the correct options and in subparts (xviii) to (xx), answer the questions as instructed.

- (i) Let R be the relation on the set $\mathbb{R}$ of real numbers defined by $R = \{(a, b) : a \leq b^2\}$. Then R is: [1]
(Understand)
- (a) reflexive only
 - (b) symmetric only
 - (c) transitive only
 - (d) neither reflexive, nor symmetric, nor transitive

- (ii) The principal value of $\sin^{-1}(\sin \frac{3\pi}{5})$ is: (*Understand*) [1]
- (a) $\frac{3\pi}{5}$
 (b) $\frac{2\pi}{5}$
 (c) $-\frac{2\pi}{5}$
 (d) $\frac{\pi}{5}$
- (iii) A is a square matrix of order 3 with $|A| = -4$. The value of $|\text{adj}(2A)|$ is: (*Apply*) [1]
- (a) 1024
 (b) -1024
 (c) 256
 (d) 64
- (iv) The radius of a circular ink blot is increasing at the rate of 0.5 cm/s. The rate at which its area is increasing when the radius is 4 cm is: (*Apply*) [1]
- (a) $2\pi \text{ cm}^2/\text{s}$
 (b) $4\pi \text{ cm}^2/\text{s}$
 (c) $8\pi \text{ cm}^2/\text{s}$
 (d) $16\pi \text{ cm}^2/\text{s}$
- (v) **Statement I:** The function $f(x) = x^3 + 3x^2 + 3x + 7$ is strictly increasing on $\mathbb{R}$. [1]
Statement II: If a function f is strictly increasing on an interval, then $f'(x) > 0$ at every point of that interval.
 (*Understand*)
- (a) Statement I is true and Statement II is false.
 (b) Statement I is false and Statement II is true.
 (c) Both the statements are true.
 (d) Both the statements are false.
- (vi) The acute angle between the curves $y^2 = x$ and $x^2 = y$ at the point (1, 1) is: (*Apply*) [1]
- (a) $\tan^{-1}(\frac{3}{4})$
 (b) $\tan^{-1}(\frac{4}{3})$
 (c) $\frac{\pi}{2}$
 (d) $\frac{\pi}{4}$
- (vii) For the function $f(x) = x^3 - 3x^2 + 3x$, the point $x = 1$ is: (*Understand*) [1]
- (a) a point of local maximum
 (b) a point of local minimum
 (c) a stationary point which is neither a local maximum nor a local minimum
 (d) a point where f is not differentiable

- (viii) The area of the shaded region shown in the figure, bounded by the curve $y = x^2$, the line $y = 4$ and the y-axis, is given by: *(Apply)* [1]



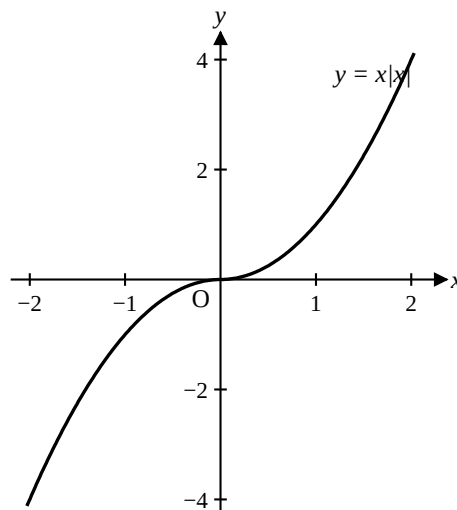
- (a) $\int_0^2 x^2 dx$
- (b) $\int_0^4 \sqrt{y} dy$
- (c) $\int_0^4 y^2 dy$
- (d) $\int_0^2 (x^2 - 4) dx$
- (ix) A student evaluates $\int \frac{1}{x \log x} dx$ as follows. [1]
- Step 1: Put $t = \log x$, so that $dt = \frac{dx}{x}$.
- Step 2: The integral becomes $\int \frac{dt}{t} = \log |t| + C$.
- Step 3: Hence the integral is $\log |x| + C$.
- In which step is there an error (if any)? *(Understand)*
- (a) Step 1
- (b) Step 2
- (c) Step 3
- (d) There is no error
- (x) Two students evaluate $\int \sin 2x dx$. Riya writes $-\frac{1}{2} \cos 2x + C$ and Karan writes $\sin^2 x + C$. [1]
- Whose answer is correct? *(Analyse)*
- (a) Only Riya's
- (b) Only Karan's
- (c) Both are correct
- (d) Neither is correct

- (xi) If $|\vec{a}| = 2$, $|\vec{b}| = 3$ and $\vec{a} \cdot \vec{b} = 3$, then $|\vec{a} \times \vec{b}|$ is equal to: (*Apply*) [1]
- (a) 3
 (b) $3\sqrt{3}$
 (c) 9
 (d) $\sqrt{3}$
- (xii) The direction cosines of the line $\frac{2x-1}{4} = \frac{3-y}{1} = \frac{z+2}{2}$ are: (*Understand*) [1]
- (a) $\frac{4}{\sqrt{21}}, \frac{1}{\sqrt{21}}, \frac{2}{\sqrt{21}}$
 (b) $\frac{2}{3}, -\frac{1}{3}, \frac{2}{3}$
 (c) $\frac{2}{3}, \frac{1}{3}, \frac{2}{3}$
 (d) $\frac{4}{\sqrt{21}}, -\frac{1}{\sqrt{21}}, \frac{2}{\sqrt{21}}$
- (xiii) The corner points of the feasible region of an LPP are (0, 2), (3, 0), (6, 0), (6, 8) and (0, 5). For $Z = 4x + 6y$, the minimum value of Z: (*Apply*) [1]
- (a) is 12 and occurs only at (0, 2)
 (b) is 12 and occurs only at (3, 0)
 (c) is 12 and occurs at every point of the segment joining (0, 2) and (3, 0)
 (d) is 0 and occurs at the origin
- (xiv) **Statement I:** If an LPP has an optimal value, it occurs at a corner point of the feasible region. [1]
Statement II: If the feasible region of an LPP is unbounded, the objective function cannot have a maximum value.
 (*Understand*)
- (a) Statement I is true and Statement II is false.
 (b) Statement I is false and Statement II is true.
 (c) Both the statements are true.
 (d) Both the statements are false.
- (xv) A and B are events with $P(A) = 0.4$, $P(B) = 0.5$ and $P(A \cup B) = 0.7$. Then $P(A' | B')$ is: [1]
 (*Analyse*)
- (a) 0.3
 (b) 0.6
 (c) 0.5
 (d) 0.4

(xvi) **Assertion:** The function $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) = x|x|$, whose graph is shown, is a bijection. [1]

Reason: A function $f : \mathbb{R} \rightarrow \mathbb{R}$ is a bijection if and only if every horizontal line meets its graph in exactly one point.

(Understand)



- (a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.
- (b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.
- (c) Assertion is true and Reason is false.
- (d) Assertion is false and Reason is true.

(xvii) **Assertion:** $\int_{-1}^1 |x| dx = 0$. [1]

Reason: $\int_{-a}^a f(x) dx = 0$ if f is an odd function.

(Understand)

- (a) Both Assertion and Reason are true and Reason is the correct explanation for Assertion.
- (b) Both Assertion and Reason are true but Reason is not the correct explanation for Assertion.
- (c) Assertion is true and Reason is false.
- (d) Assertion is false and Reason is true.

(xviii) If the matrix $A = \begin{bmatrix} 0 & a & 3 \\ 2 & 0 & -1 \\ b & 1 & 0 \end{bmatrix}$ is skew-symmetric, find the values of a and b . (Understand) [1]

(xix) Find the sum of the order and the degree of the differential equation $[1 + (\frac{dy}{dx})^2]^{3/2} = k \frac{d^2y}{dx^2}$. [1]

(Analyse)

(xx) The probability distribution of a random variable X is $P(X = x) = kx$ for $x = 1, 2, 3, 4$ (and 0 otherwise). Find $E(X)$. (Apply) [1]

SECTION B – 14 MARKS

Question 2

Find the angle between the line $\frac{x+1}{2} = \frac{y}{3} = \frac{z-3}{6}$ and the plane $10x + 2y - 11z = 3$. [2]

(Apply)

Question 3

A problem in Mathematics is given to three students A, B and C, whose chances of solving it are $\frac{1}{2}$, $\frac{1}{3}$ and $\frac{1}{4}$ respectively. They try independently.

- (i) Find the probability that the problem is solved. (Apply) [1]
(ii) Find the probability that exactly one of them solves it. (Evaluate) [1]

Question 4

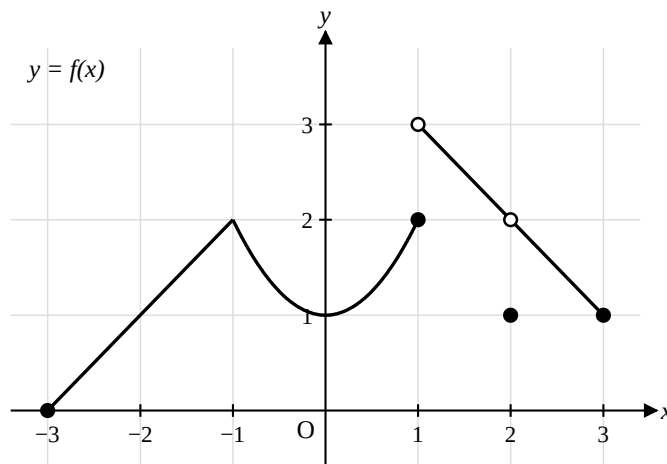
Solve the differential equation $2xy \frac{dy}{dx} = x^2 + y^2$. (Apply) [2]

Question 5

Find the values of a and b so that the function $f(x) = \begin{cases} ax^2 + 1, & x \leq 1 \\ x^2 + bx + 3, & x > 1 \end{cases}$ is differentiable at $x = 1$. (Understand) [2]

OR

The graph of a function f defined on $[-3, 3]$ is shown.



- (a) At which point does f have a removable discontinuity? Give a reason. (Understand) [1]
(b) At which point in $(-3, 3)$ is f continuous but not differentiable? Give a reason. (Analyse) [1]

Question 6

Solve for x: $\tan^{-1}2x + \tan^{-1}3x = \frac{\pi}{4}$. (Apply) [2]

OR

Prove that $\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{8} = \frac{\pi}{4}$. (Apply) [2]

Question 7

Find λ if the scalar projection of $\vec{a} = \lambda\hat{i} + 2\hat{j} + \hat{k}$ on $\vec{b} = 2\hat{i} - 2\hat{j} + \hat{k}$ is 3. Hence write the vector projection of $\vec{a}$ on $\vec{b}$. (Apply) [2]

OR

Find the area of the parallelogram whose diagonals are represented by the vectors $2\hat{i} - \hat{j} + \hat{k}$ and $3\hat{i} + 4\hat{j} - \hat{k}$. (Apply) [2]

Question 8

A cable car runs along a straight cable held up by pylons. With respect to a fixed origin at the base station, three pylons are at P(1, 2, 3), Q(3, 4, 7) and R(4, 5, 9) (distances in hundreds of metres).

- (i) Using vectors, show that the three pylons are collinear. (Understand) [1]
- (ii) Find the ratio in which Q divides PR. (Apply) [1]

SECTION C – 21 MARKS**Question 9**

A small workshop makes wooden toy cars and toy trains. Each car needs 2 hours of cutting and 1 hour of finishing; each train needs 1 hour of cutting and 3 hours of finishing. On a day at most 10 hours of cutting and 15 hours of finishing are available. The profit is ₹30 on a car and ₹20 on a train, and everything made is sold.

- (i) Formulate the problem as a linear programming problem to maximise the profit. (Apply) [1]
- (ii) Draw the feasible region and find its corner points. (Apply) [1]
- (iii) Find the number of cars and trains that maximise the profit, and the maximum profit. (Analyse) [1]

Question 10

If $y = \frac{\sin^{-1}x}{\sqrt{1-x^2}}$, prove that $(1-x^2) \frac{d^2y}{dx^2} - 3x \frac{dy}{dx} - y = 0$. (Apply) [3]

Question 11

Let $f(x) = \tan^{-1}\left(\frac{\sqrt{1+x^2}-1}{x}\right)$, $x \neq 0$.

- (i) Using a suitable substitution, show that $f'(x) = \frac{1}{2(1+x^2)}$. (Apply) [2]
- (ii) A student claims that $f(x)$ increases without bound as x becomes large. Evaluate the claim. (Evaluate) [1]

Question 12

Find the shortest distance between the lines $\vec{r} = (\hat{i} + 2\hat{j} + 3\hat{k}) + \lambda(\hat{i} - 3\hat{j} + 2\hat{k})$ and $\vec{r} = (4\hat{i} + 5\hat{j} + 6\hat{k}) + \mu(2\hat{i} + 3\hat{j} + \hat{k})$. (Apply) [3]

OR

P(1, 6, 3) is a point and L is the line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$.

- (a) Write the coordinates of a general point Q of L and the direction ratios of PQ. (Understand) [1]
- (b) Find the foot of the perpendicular from P to L. (Apply) [1]

- (c) Find the perpendicular distance of P from L. (*Apply*) [1]

Question 13

Using properties of determinants, prove that $\begin{vmatrix} a & b & c \\ a^2 & b^2 & c^2 \\ bc & ca & ab \end{vmatrix} = (a-b)(b-c)(c-a)(ab+bc+ca).$ [3]

(*Analyse*)

OR

Two families A and B have the following members: A has 4 men, 6 women and 2 children; B has 2 men, 2 women and 4 children. The recommended daily requirement of calories is 2400 for a man, 1900 for a woman and 1800 for a child, and of proteins is 45 g for a man, 55 g for a woman and 33 g for a child. Using matrix multiplication, find the total requirement of calories and proteins for each family. Which family needs more protein per member? (*Analyse*) [3]

Question 14

Answer the following.

- (i) Prove that $\int_a^b f(x) dx = \int_a^b f(a+b-x) dx.$ (*Understand*) [1]

- (ii) Hence evaluate $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{\sqrt{\sin x}}{\sqrt{\sin x} + \sqrt{\cos x}} dx.$ (*Apply*) [2]

Question 15

Consider the parabolas $y^2 = 4x$ and $x^2 = 4y.$

- (a) Find their points of intersection. (*Understand*) [1]
 (b) Using integration, find the area of the region enclosed between them. (*Apply*) [2]

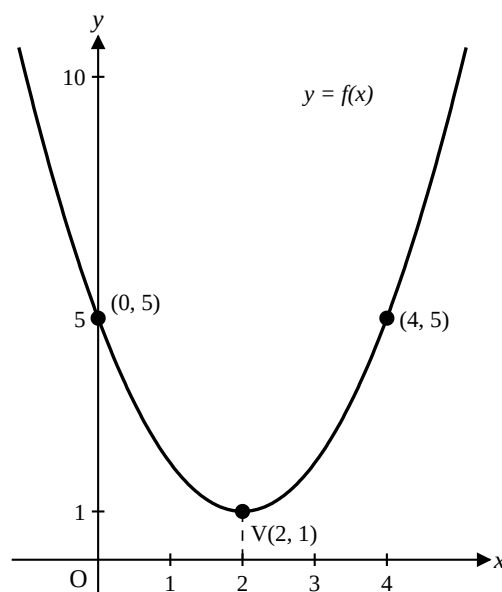
OR

Using integration, find the area of the region in the first quadrant enclosed by the x-axis, the line $y = x$ and the circle $x^2 + y^2 = 16.$ (*Apply*) [3]

SECTION D – 25 MARKS

Question 16

The height (in metres) of a small hill along a straight path is modelled by $f(x) = x^2 - 4x + 5$, $x \in \mathbb{R}$, whose graph is shown.



- (i) Find the range of f . (*Understand*) [1]
- (ii) Is $f : \mathbb{R} \rightarrow \mathbb{R}$ invertible? Justify. (*Analyse*) [1]
- (iii) $g : [2, \infty) \rightarrow [1, \infty)$ is defined by $g(x) = f(x)$. Find $g^{-1}(x)$. (*Apply*) [1]
- (iv) $h : (-\infty, 2] \rightarrow [1, \infty)$ is defined by $h(x) = f(x)$. Find $h^{-1}(x)$ and verify that $(h \circ h^{-1})(5) = 5$. (*Apply*) [1]
- (v) Find the value of x for which $g^{-1}(x) = h^{-1}(x)$, and interpret it on the graph. (*Analyse*) [1]

Question 17

A right circular cone is inscribed in a sphere of radius 9 cm, its vertex and the circumference of its base lying on the sphere. Let the height of the cone be h cm and the radius of its base r cm.

- (a) Show that $r^2 = 18h - h^2$. (*Understand*) [1]
- (b) Find h for which the volume of the cone is maximum, and justify that it is a maximum. (*Apply*) [2]
- (c) Find the maximum volume of the cone. (*Apply*) [1]
- (d) Show that the maximum volume of the cone is $\frac{8}{27}$ of the volume of the sphere. (*Evaluate*) [1]

OR

A ladder 5 m long leans against a vertical wall. Its foot is pulled along the level ground away from the wall at 2 m/s. Let x m be the distance of the foot from the wall, y m the height of the top, and θ the angle the ladder makes with the ground.

- (a) How fast is the top sliding down when the foot is 4 m from the wall? (*Apply*) [2]
- (b) Find the rate of change of the area of the triangle formed by the ladder, the wall and the ground at that instant. Is the area increasing or decreasing? (*Analyse*) [2]
- (c) Find the rate at which θ is changing at that instant. (*Apply*) [1]

Question 18

Answer the following.

- (a) Using partial fractions, evaluate $\int \frac{1}{x(x^2 + 1)} dx$. (*Understand*) [2]
- (b) Hence, evaluate $\int \frac{1}{x(x^4 + 1)} dx$. (*Evaluate*) [3]

OR

- (a) Prove that $\int e^x [f(x) + f'(x)] dx = e^x f(x) + C$. (*Understand*) [2]
- (b) Hence, evaluate $\int_0^1 e^x \frac{x-1}{(x+1)^3} dx$. (*Evaluate*) [3]

Question 19

In a factory, machines E_1 , E_2 and E_3 produce 50%, 25% and 25% of the bulbs respectively. Of their output, 4%, 4% and 5% respectively are defective. A bulb is picked at random from the day's production.

- (a) Find the probability that the bulb is defective. (*Apply*) [2]
- (b) The bulb is found to be defective. Find the probability that it was made by E_1 . (*Apply*) [2]
- (c) The quality inspector says that a defective bulb most likely came from E_3 , because E_3 has the highest defect rate. Is he right? Justify. (*Evaluate*) [1]

OR

A disease affects 1% of a population. A test detects the disease in 99% of people who have it, but also gives a positive result for 2% of people who do not have it. A person is tested at random.

- (a) Find the probability that the result is positive. (*Apply*) [2]
- (b) The result is positive. Find the probability that the person has the disease. (*Analyse*) [2]
- (c) Comment on the reliability of a single positive result. (*Evaluate*) [1]

Question 20

A person invests a total of ₹50,000 in three schemes paying simple interest at 6%, 8% and 9% per annum. The total annual interest is ₹4,000. The amount invested at 9% is ₹10,000 less than the combined amount in the other two schemes. Form a matrix equation and solve it, using the inverse of the coefficient matrix, to find the amount in each scheme. (*Evaluate*) [5]

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